



PARKTOWN BOYS' HIGH SCHOOL
DEPARTMENT OF GEOGRAPHY

GEOGRAPHY

—PRELIM PAPER 1—

6 September 2024

MEMO

Examiner: Mr. A Meintjes
Internal Moderator: Mr. D. Grace
External Moderator:

TIME: 3 hours

MARKS: 150 Marks

SECTION A – Climate and Weather

QUESTION 1: Short Questions

1.1. Various possible answers are provided for each question. Write down **only the letter** of the correct answer next to the corresponding number.

1.1.1. Pressure belts can be defined as ...

- A. A localised area of low pressure such as a cold front.
- B. Areas of differing pressure that extend around the planet going from East to West.
- C. Extended areas of differing pressure found along the coastline.
- D. Areas of differing pressure that extend around the planet going from North to South. (1)

1.1.2. A North facing slope in South Africa would experience ...

- A. Warmer conditions between December and February.
- B. Colder conditions between December and February.
- C. Consistently warmer conditions throughout the year.
- D. Consistently cooler conditions throughout the year. (1)

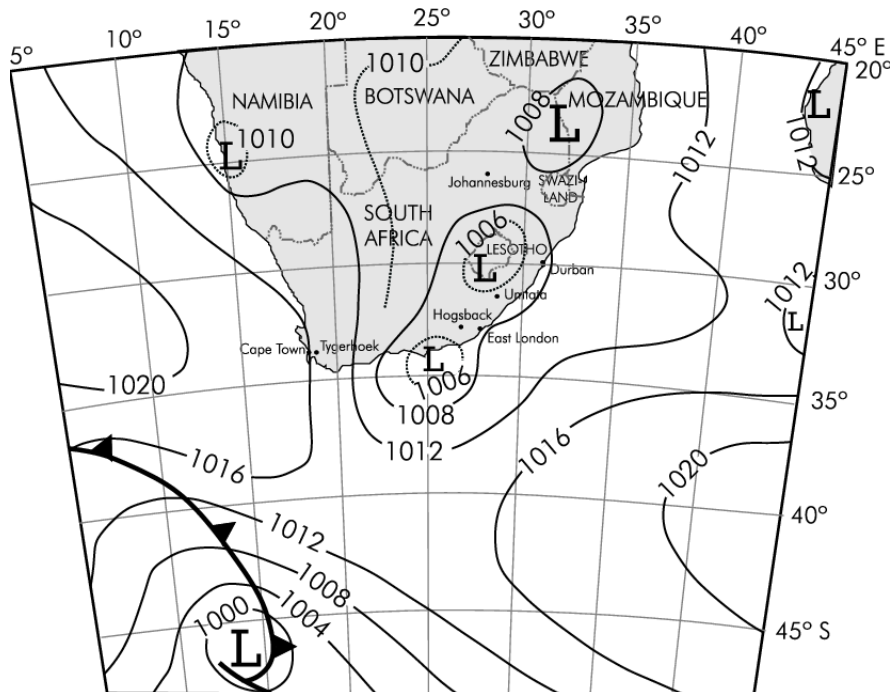
1.1.3. Katabatic winds refer to ...

- A. Winds that blow up valley sides and along the valley floor during the day.
- B. Winds that travel vertically up in cities due to excess heat sources.
- C. Sinking winds resulting from temperature inversions.
- D. Winds that blow down valley sides and along the valley floor during the night. (1)

1.1.4. Anabatic winds refer to ...

- A. Winds that blow up valley sides and along the valley floor during the day.
- B. Winds that travel vertically up in cities due to excess heat sources.
- C. Sinking winds resulting from temperature inversions.
- D. Winds that blow down valley sides and along the valley floor during the night. (1)

Refer to the diagram below showing a typical synoptic weather map and answer the questions that follow.



1.1.5. The season shown above is most likely ...

- A. Summer.
- B. Autumn.
- C. Winter.
- D. Spring.

(1)

1.1.6. The weather phenomenon seen approaching South Africa from the bottom left is a ...

- A. Mid-latitude cyclone.
- B. Hurricane.
- C. Tropical cyclone.
- D. Cut-off low.

(1)

1.1.7. The low pressure cells that are dominating the interior of South Africa have developed because ...

- i. There is an increase in heat this time of the year.
- ii. The Kalahari High Pressure has lifted.
- iii. The interior is hotter than the coastline.
- iv. The cold Benguela current has weakened.

A. i & iii ONLY.

B. None of the above.

C. All of the above.

D. i, ii & iii ONLY.

(1)

1.1.8. Low pressure systems in Summer can lead to ...

A. Colder conditions over a few weeks.

B. Clearer skies.

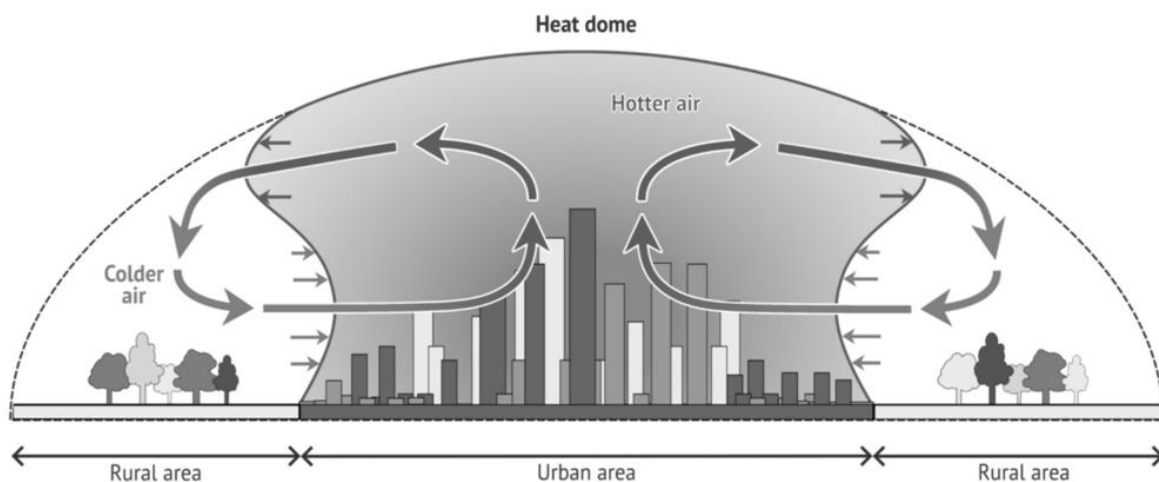
C. Increased flooding risks.

D. Increased temperature over a few days.

(1)

(8 x 1) (8)

1.2. Refer to the diagram below and answer the questions that follow. For each answer select the correct word in brackets. Write only the question and the correct word. E.g. 1.2.8 Jam.



- 1.2.1. Air rising in the urban area would signify an area of (high pressure/low pressure).
- 1.2.2. Rural areas are cooler due to having (increased wind/more vegetation) than urban areas.
- 1.2.3. One reason for the air being warmer over urban areas is (an increase of artificial heating/more exposure to sunlight).
- 1.2.4. Due to the urban heat island effect in summer, urban areas receive (more rainstorms/less rainstorms).
- 1.2.5. The typical type of rain experienced in urban areas as a result of this heat island effect is (orographic/convectional) rain.
- 1.2.6. In winter the urban heat island effect is subdued. This is because there is a lack of (artificial heat sources/sunlight) contributing to the effect.
- 1.2.7. In winter we see the creation of pollution domes. Pollution is trapped over urban cities by (an inversion layer/cold fronts).

(7 x 1) (7)

[15]

Question 1		Question 2	
1	B	1	Low pressure
2	A	2	More veg
3	D	3	Artificial heating
4	A	4	More rainstorms
5	A	5	Convectional
6	A	6	Sunlight
7	D	7	Inversion layer
8	C		

QUESTION 2: Tropical Cyclones

2. Refer to the article below and answer the questions that follow.

New forecast reveals when and where hurricane season could get active after Ernesto

By [Mary Gilbert](#), CNN Meteorologist

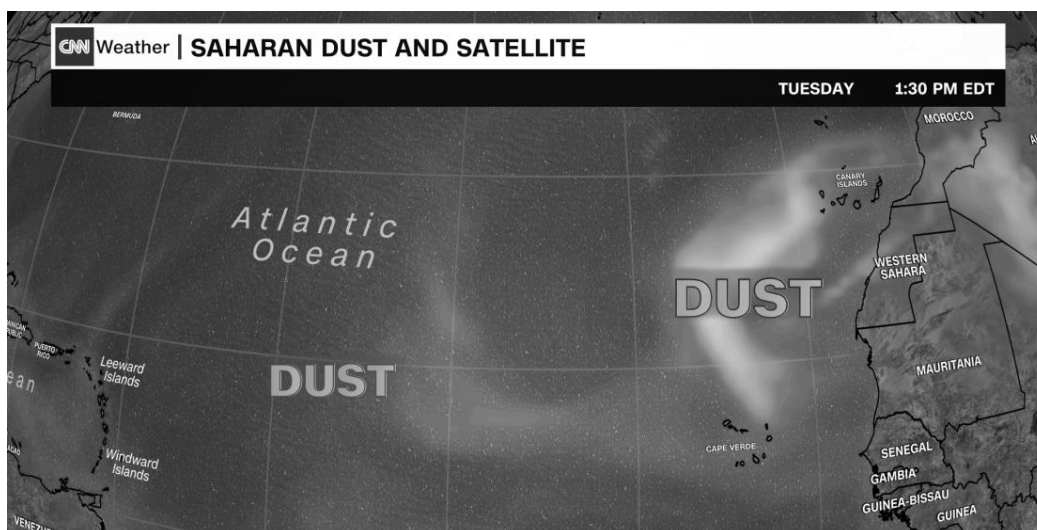
Published 3:19 PM EDT, Tue August 20, 2024

CNN —

Tropical activity in an unusually active hurricane season will take a breather after [Ernesto](#), but forecasters don't believe it will last long.

Atlantic storms will be limited for at least the next week. Dry, dusty air and some disruptive winds in the upper atmosphere are making it difficult for stormy weather to get its act together. A [similar setup](#) kept the Atlantic quiet for a few weeks following record-breaking Hurricane [Beryl](#).

But this brief break in activity isn't likely to last nearly as long, according to forecasts from the Climate Prediction Center. Once the dry, dusty air eases, Atlantic tropical activity could be off to the races as stormy weather pushes off the coast of Africa and out to sea unimpeded.



A plume of dust impacts part of the Atlantic Ocean on Tuesday afternoon.

This isn't surprising because the climatological peak of hurricane season is just a few weeks away.

The Atlantic ocean is also still near record-warm, which could help systems develop and potentially [explode in strength](#) – something becoming more common in a world warming due to fossil fuel pollution.

The season has already been unusually active, producing five named storms – three of which became hurricanes – by mid-August. Ernesto strengthened into the third hurricane of the season [almost a month earlier](#) than normal, according to the NHC.

Beryl also shattered expectations when it became the earliest Category 5 hurricane on record in early July. It was a major hurricane – Category 3 or stronger – almost two months ahead of schedule.

This season has already racked up about half of the activity an entire normal season would produce with plenty of tropical activity to come.

2.1. What ocean did hurricane Ernesto develop over? (1 x 1) (1)

Atlantic ✓

2.2. How many hurricanes have developed over the Atlantic this year, including Ernesto? (1 x 2) (2)

5 ✓✓

2.3. According to the article, dry and dusty air is causing a 'break' in hurricane season. Why would dry and dusty air cause this 'break'? (1 x 2) (2)

[Any]

Hurricanes need moisture ✓✓

Dry air reduces moisture ✓✓

2.4. Hurricanes develop over the Atlantic Ocean. What is the name we give to these massive storms when they develop over the Indian Ocean? (1 x 1) (1)

Tropical Cyclones ✓

2.5. When considering hurricane Ernesto, quote ONE piece of evidence from the article to support the notion that climate change may be responsible for these storms developing so early. (1 x 1) (1)

"The season has already been unusually active, producing five named storms – three of which became hurricanes – by mid-August." ✓

"Ernesto strengthened into the third hurricane of the season [almost a month earlier](#) than normal" ✓

NOTE: CANNOT QUOTE RE BERYL, QUESTION ASKS FOR ERNESTO

- 2.6. In a paragraph of approximately EIGHT lines, discuss the social impacts of hurricanes and the strategies we can implement to mitigate the damage caused by these phenomena.

(4 x 2) (8)

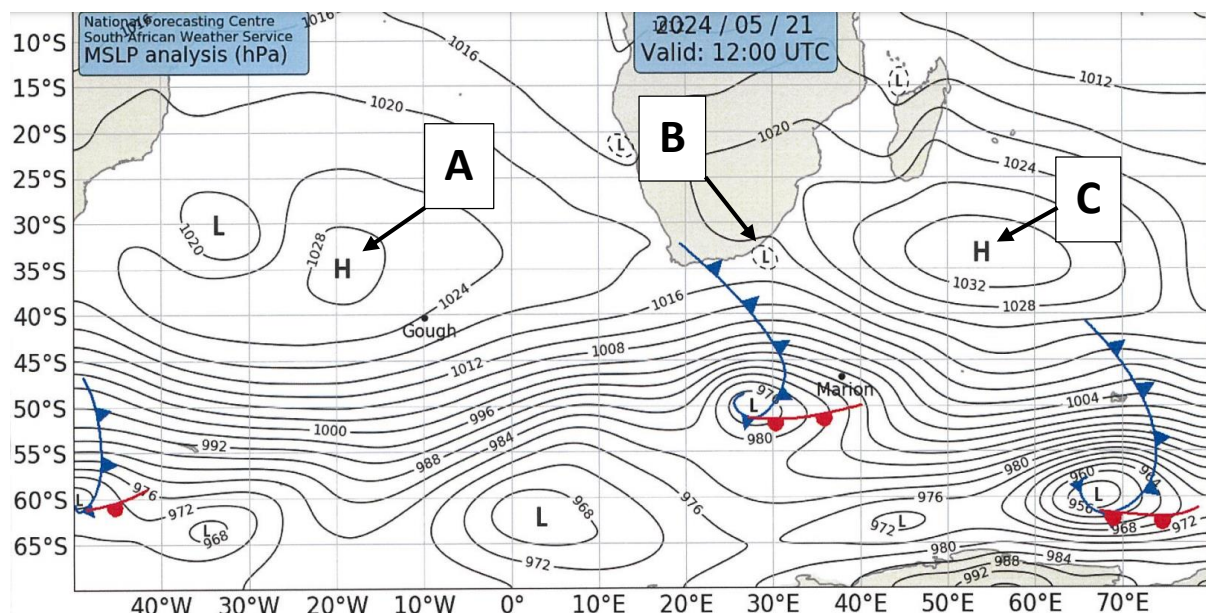
[Accept reasonable responses. NOTE: can have 3 impacts and 1 strat or visa versa. MUST have both for full marks though]

Social impacts – Need to be discussion Jobs lost Deaths in family Family structure ruined Trauma Breaks in education Etc. ✓✓✓✓	Strategies – Needs to be a discussion Planning Prediction and warning Community preparations Manage the loss Etc. ✓✓✓✓
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[15]

QUESTION 3: Anti-cyclones

3. Study the synoptic weather map below and answer the questions that follow.



- 3.1. What season is this synoptic weather map depicting?

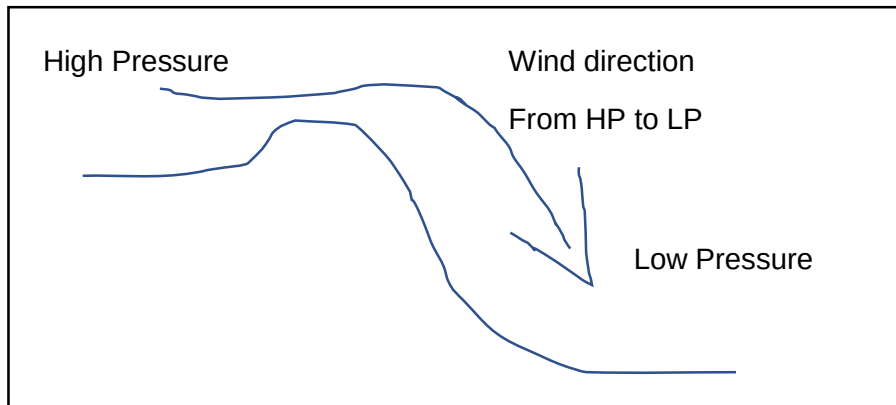
(1 x 1) (1)

Winter ✓

- 3.2. Provide the names for the pressure systems labelled A and C on the map respectively. (2 x 1) (2)

South Atlantic and South Indian HP/Anticyclone (MUST HAVE SOUTH) ✓✓

- 3.3. Draw a simple cross section showing how the low pressure named at B may lead to the formation of Berg Winds. (4 x 1) (4)



✓✓ - High and low pressure labels

✓✓ - Wind direction

- 3.4. Discuss why Berg Winds are a danger to South Africa. Provide TWO reasons in your answer (2 x 2) (4)

[Accept reasonable answers – MUST BE A DISCUSSION]

Dry out crops and vegetation ✓✓

Exacerbate veld fires ✓✓

Disrupt ecosystems ✓✓

Can cause heat stroke ✓✓

Increases electricity usage (air cons etc.) ✓✓

Fires can damage infrastructure ✓✓

- 3.5. Not shown on this synoptic weather map is the Kalahari High Pressure cell that dominates the interior of South Africa during this season. Define the term *High Pressure*. (1 x 2) (2)

[Concept]

Descending, heavy air ✓✓

- 3.6. Describe the role that the Kalahari High Pressure cell plays in the reduction of rain over the interior during this season. (1 x 2) (2)

Prevents the rising of air ✓✓

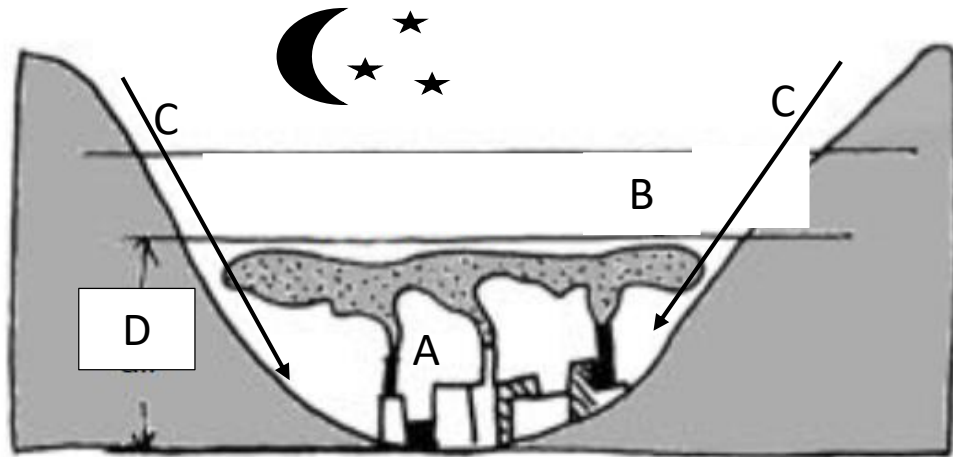
Prevents air condensing ✓✓

Prevents cloud formation ✓✓

[15]

QUESTION 4: Valley Climates

4. Study the figure below showing a typical valley climate and answer the questions that follow.



- 4.1. Name the wind that flows down a slope at night, labelled C. (1 x 1) (1)

Downslope wind/Katabatic wind ✓

- 4.2. Explain how this wind formed. (2 x 2) (4)

Air temperatures drop ✓✓

Cold air is heavy and sinks ✓✓

High pressures above causes air to descend ✓✓

At night there is a lack of heat sources (no rising air at night) ✓✓

- 4.3. Indicate if warm or cold air would be found at A and B respectively. (2 x 1) (2)

A – Cold B – Warmer ✓✓

- 4.4. Fully explain how the inversion layer at B would form. (3 x 2) (6)

[Accept reasonable answers]

Clear cloudless skies result in heavy descending air ✓✓

This cool air sinks down towards the valley floor ✓✓

Cold air begins to collect on the valley floor, displacing all warm air ✓✓

This warm air rises to form a thermal belt, resulting in a temperature inversion ✓✓

- 4.5. List TWO potential concerns for people settling on the valley floor. (2 x 1) (2)

Frost pockets may kill veg/crops ✓✓

Increase costs, Cold conditions are dangerous etc. ✓✓

[15]

TOTAL SECTION A: [60]

SECTION B - Geomorphology

QUESTION 5: Short Questions

5.1. The following descriptions are all features of a river system. State whether each of these is found in the UPPER, MIDDLE or LOWER course of a river.
E.g. 5.1.8. Upper

5.1.1. Erosion is dominant at in the lower course of a river.

5.1.2. In an ideal transverse river profile, erosion and deposition would be equal.

5.1.3. Rapids and waterfalls are a features of the upper river course.

5.1.4. Most headward erosion takes place in the middle course of a river.

5.1.5. Oxbow lakes typically form within deltas.

5.1.6. The Nile travelling through the desert would be an example of an episodic river.

5.1.7. Saltation refers to the percentage of salt in a river.

(7 x 1) (7)

5.2. Match the terms in the table with one of the descriptions. Write only the number of the question and the correct term (5.2.1–5.2.8) in the ANSWER BOOK, for example 5.2.9 Sunshine.

Capacity	Knickpoint	Sinuosity
Valley	Meander	Abrasion
Antecedent drainage	Hydraulic action	Source
Watershed	Confluence	Interfluves
Mouth	Laminar flow	Turbulent flow

5.2.1. The erosion of the river bottom and the riverbank by the material carried in the river.

5.2.2. The water level, or stage, at which a stream, river or lake is at the top of its banks and any further rise would result in water moving into the flood plain.

5.2.3. A ridge or area of land dividing two river valleys within a drainage basin.

5.2.4. A term to describe a location in a river or channel where there is a sharp change in channel slope.

5.2.5. A process whereby the continual colliding of water and other materials on the base of the plunge pool gradually deepens the pool.

5.2.6. The term that refers to the beginning of a river.

5.2.7. An elevated area of land that separates two drainage basins.

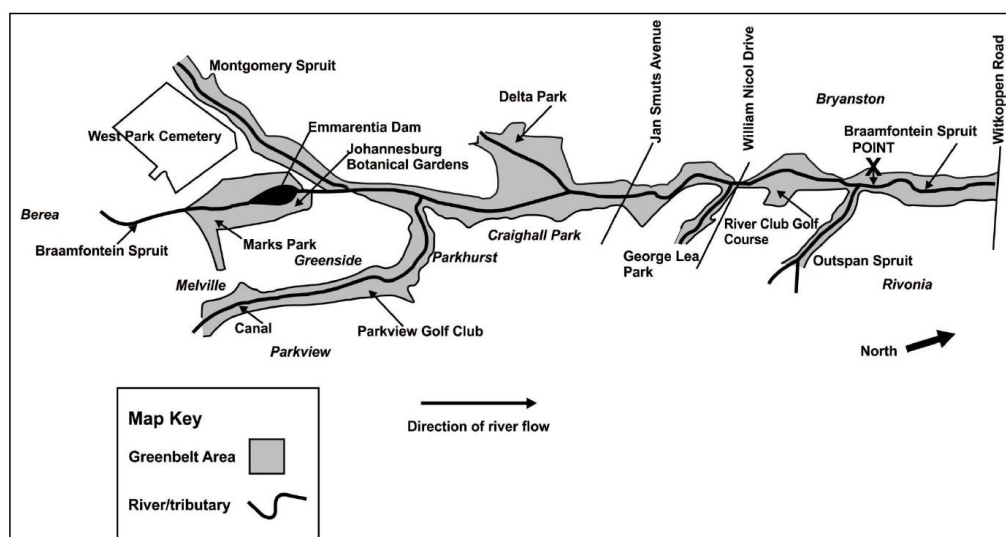
5.2.8. A term used to describe calm water with little turbulence.

(8 x 1) (8)
[15]

Question 5.1		Question 5.2	
NOTE: Question error – Should have been true or false			
1	False (upper)	1	Abrasion
2	True (middle)	2	Capacity
3	True (upper)	3	Interfluves
4	False (upper)	4	Knickpoint
5	False (lower)	5	Hydraulic Action
6	False (middle)	6	Source
7	False (...)	7	Watershed
		8	Laminar flow

QUESTION 6: Drainage basins

6. Refer to the figure below showing a drainage basin and answer the questions that follow.



[Source: <<http://www.amethyst.co.za/Veld/BraamfonteinSpruitBryanston>>]

6.1. **Name** TWO tributaries of the Braamfontein Spruit? (2 x 1) (2)

Montgomery Spruit

Delta Park

Canal

Outspan Spruit ✓✓

6.2. Using only the figure above, **determine** the stream order of the Braamfontein Spruit when the river passes Witkoppen Road. (1 x 2) (2)

2nd Order ✓✓

6.3. **Name** ONE temporary base level along the Braamfontein Spruit as seen above. (1 x 1) (1)

Emmerentia Dam ✓

6.4. With reference to the source above, **suggest** THREE ways in which urban development has impacted upon the natural channel flow of the Braamfontein Spruit. (3 x 2) (6)

[Accept reasonable answers – MUST LINK TO CHANNEL FLOW]

We have increased pollution (slowed flow down) ✓✓

Jan Smuts/William Nicol – Construction disrupts flow ✓✓

Golf courses – Dammed up water disrupts flow ✓✓

Emerentia Dam – Stopped flow ✓✓

The canal may have increased flow ✓✓

6.5. Storm water drains in the Braamfontein Spruit catchment are often blocked with rubbish. **Suggest** how this may impact upon:
The **discharge** of the Braamfontein Spruit and
The **health** of the various animals, birds and fish found in this area. (2 x 2) (4)

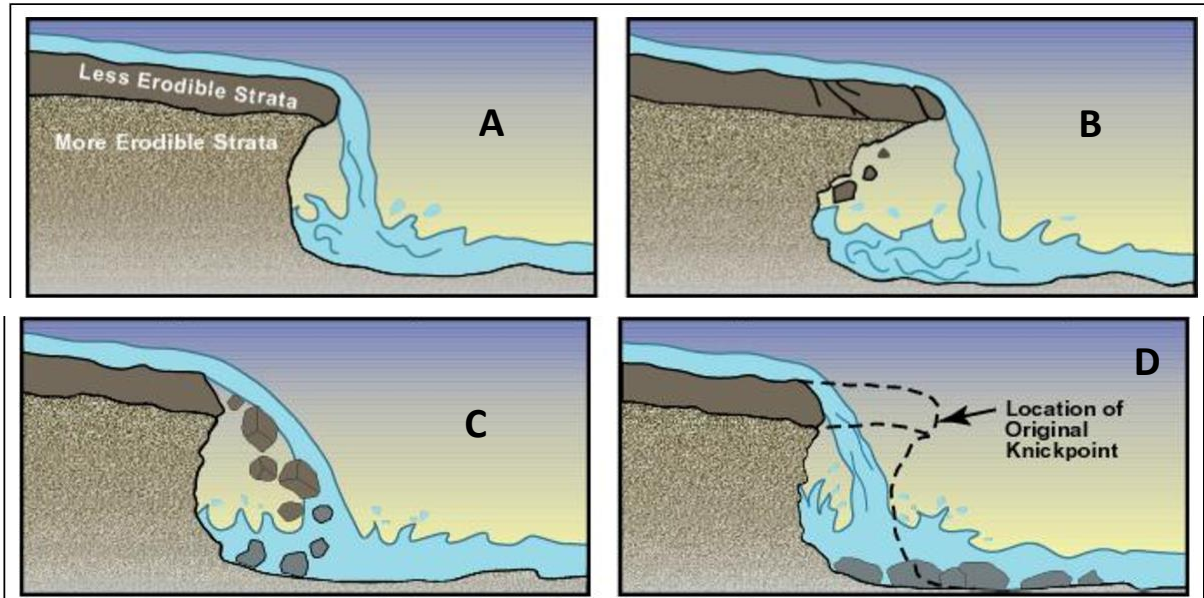
Slow down the discharge/block up the river ✓✓

Reduce the health/kill all the animals, birds etc. ✓✓

[15]

QUESTION 7: Fluvial Processes

7. Study the figure below and answer the questions that follow.



7.1. What feature is being shown in the diagram above? (1 x 1) (1)

Waterfall ✓

7.2. What is another term for “Less Erodible Strata”? (1 x 1) (1)

Resistant strata, hard rock etc. ✓

7.3. Name the erosive process occurring at B. (1 x 1) (1)

Undercutting ✓

7.4. The feature identified in QUESTION 7.1. is migrating headwards. By referring to the diagram, explain how this process works. (3 x 2) (6)

[Any 3 points get marks]

Increased erosive power of the water removes rock (can mention rejuvenation) ✓✓

Soft rock (more erodible rock) is weather away ✓✓

Undercutting occurs removing soft rock ✓✓

Hydraulic action and abrasion removes soft rock ✓✓

Resistant rock above eventually collapses ✓✓

This is removed exposing more soft rock and creating a new lip ✓✓

Process repeats as long and the waterfall heads towards the source ✓✓

7.5. Features like the above often develop as a result of river rejuvenation.

Define the term *River Rejuvenation*.

(1 x 2) (2)

[Concept]

Increase in erosive power of a river ✓✓

Increase in water volumes and power of a river ✓✓

7.6. Describe TWO negative consequences for people and/or animals that live downstream of a river that has been rejuvenated.

(2 x 2) (4)

[Accept reasonable answers]

Increased flood risk ✓✓

Possible infrastructure destruction ✓✓

Possible danger and death from flooding ✓✓

Livelihoods and crops destroyed etc. ✓✓

[15]

QUESTION 8: River Management

Study the article below and answer the questions that follow.

SA's rivers and dams can no longer recover from pollution, say water scientists

The National Water Act has failed to meet its objective of improving the quality of water in South Africa's rivers and dams, according to a resource management scientist.

"We release about seven billion litres of sewage every single day into our rivers and dams and the purpose of the National Water Act at its heart is to improve the quality of water resources," said Anthony Turton, a professor at the Centre for Environmental Management at the University of the Free State.

"Many of our rivers and dams have now reached a point of collapse where they can no longer recover so we can state with empirical evidence that the National Water Act has failed; it's failed to meet its objective of improving the quality of water in the rivers and dams."

He was speaking at a recent Tipping Points webinar on “South Africa’s Water Whirlpool of Pain”, hosted by the Oppenheimer Generations Research and Conservation. There, panellists discussed how the country’s water crisis is accelerating beyond its capacity to fix it.

About 80% of the sewage released into waterways comes back into the system either partially treated or, in some cases, untreated. “How long can we continue without seeing human health implications? It’s only a question of time,” he said, referring to the 30 cholera deaths in Hammanskraal last year.

Sewage does not only contain human waste but also pathogens such as bacteria and viruses, as well as partially metabolised medication. This includes high levels of oestrogen for birth control purposes and partially metabolised antiretroviral medication partially metabolised antibiotics. “It’s not inconceivable that multidrug-resistant pathogens will come out of these places.

“We have some evidence now — there’s probably eight to 10 cases that I’m aware of — where people have been in contact with bacteria in the river where they have either injured themselves slightly ... and that has created an open wound in which a certain bacteria has entered the body: flesh-eating bacteria.”

According to Turton, the rate of pollution in rivers and the rate of change in these river systems is “consistent with this notion of a tipping point”.

“The change in the country’s rivers is becoming so rapid that I believe there’s a strong possibility that it will start exceeding our institutional capacity to cope.”

Yet at the same time, public confidence and public trust in the government and the institutions that provide water is under severe pressure. “We need to go out of our way to have communication that is honest and accurate and to have incredible leadership.”

8.1. What was the objective of the National Water Act? (1 × 1) (1)

Improve water quality ✓

8.2. According to the article, why did the National Water Act fail? (1 × 2) (2)

Pollution levels have increased in our water system ✓✓

Leadership has failed ✓✓

Maintenance has failed ✓✓

8.3. What percentage of sewerage released into rivers re-enters the water system? (1 x 1) (1)

80% ✓

8.4. What does EIA stand for? (1 x 1) (1)

Environmental Impact Assessment ✓

8.5. Describe ONE way that EIAs may help improve the quality of our rivers. (1 x 2) (2)

[Accept reasonable responses]

Research may help us plan clean-ups ✓✓

Fines can be given to people that are contributing negatively to the problem ✓✓

Officials can be held accountable ✓✓

Resources can be correctly allocated to the most vulnerable areas ✓✓

Etc.

8.6. Considering the following quote from the article: "We need to go out of our way to have communication that is honest and accurate and to have incredible leadership." Write a short paragraph in which you explore what the future may hold for South Africans if we do not improve our current situation, and provide solutions that leadership could consider to improve the situation. (4 x 2) (8)

[Accept reasonable answers. NOTE: Answers may have 3 future consequences and 1 solution or visa versa. Need to have both for full marks]

Future consequences	Solutions
Increases in various diseases	EIAs
Increased cost to health sector	Fine offenders
Increased cost to water purification	Legislature change
Loss of tourism	Restrict development along rivers
Loss of crops	Etc. ✓✓✓✓
Etc. ✓✓✓✓	

[15]

TOTAL SECTION B: [60]

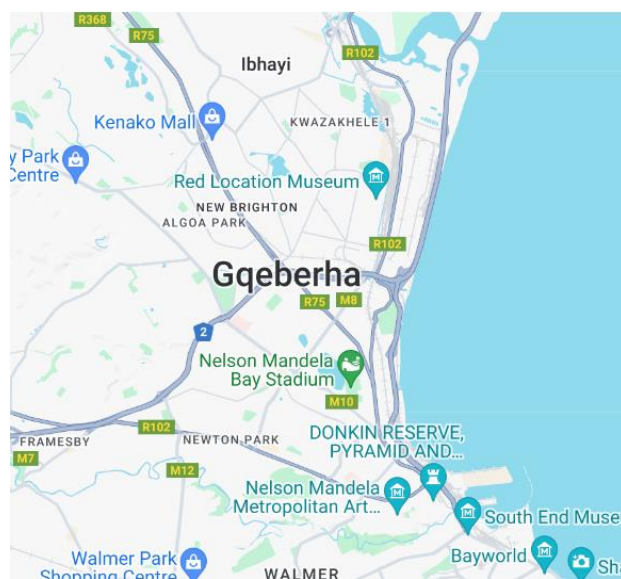
SECTION C - Mapwork

RESOURCE MATERIAL

1. 3325DC & DD and 3425BA GQEBERHA and an orthophoto 3325 DC 23

GQEBERHA

GENERAL INFORMATION ON GQEBERHA



Gqeberha (Xhosa: [ǀǀʰɛbɛːxà]; English: /kɛˈbɛərɪə/ *keh-BAIR-khə*), formerly known as **Port Elizabeth**, and colloquially referred to as **P.E.**, is a major seaport and the most populous city in the Eastern Cape province of South Africa. It is the seat of the Nelson Mandela Bay Metropolitan Municipality and home of the Nelson Mandela Bay Stadium. It is also South Africa's second-smallest metropolitan municipality by area. It is the sixth-most populous city in South Africa and is the cultural, economic and financial hub of the Eastern Cape.

Located on the western portion of Algoa Bay along the southeastern coast of South Africa, the city lies 770 km east of Cape Town. It is east of the Garden Route and faces the Indian Ocean. It covers 251 square kilometers of the Nelson Mandela Bay metropolitan area, and is administered by South Africa's sixth-largest metropolitan municipality. The city's warm oceanic climate ranks it among the top cities in the world for pleasant year-round weather. The city is known for many blue-flag beaches along the city's urban coastline; its popularity as an international and local holiday destination; and its rich and diverse cultural heritage. It is a gateway city for the Eastern Cape's adventure, outdoor and African big five game safari tourism.

QUESTION 9: Calculations

- 9.1. What is the name of the stadium located at F on the topographical map? (1 x 1) (1)

Nelson Mandela Stadium ✓

- 9.2. Provide the coordinates for the Trig Beacon 266 in block D3. (2 x 1) (2)
- ✓✓

- 9.3. Provide the map code or map reference for the map North of 3325DC. (1 x 1) (1)

3325DA ✓

- 9.4. Choose the correct option below.

The area covered by the orthophoto map as represented on the topographic map is approximately:

B. 3.885km². ✓✓

(1 x 2) (2)

- 9.5. Calculate the current magnetic declination for 2024. Show ALL calculations. Use the following headings as a guide:

The difference in years: ✓

Mean annual change: ✓

Total change: ✓

Magnetic declination for 2024: ✓

(4 x 1) (4)

[10]

QUESTION 10: Interpretation

- 10.1. The Bakensrivier exists the drainage basin at block C5.

- 10.1.1. What is the term given to the part of the river where it meets the ocean? (1 x 1) (1)

Mouth ✓

10.1.2. In what course would this river be, middle course or low course? (1 x 1) (1)

Lower course ✓

10.1.3. This river system has a medium drainage density. Discuss TWO factors that would influence this system's drainage density. (2 x 2) (4)

Climate – More rain = more density ✓✓

Rock type – Hard rock = More density ✓✓

Infiltration/soil – More permeable = less density ✓✓

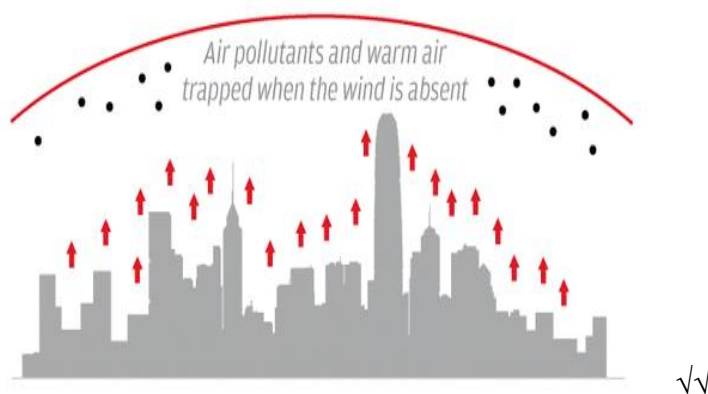
10.2. Refer to the vast amount of industrialisation in this area (H).

10.2.1. How may this industrialisation lead to acid rain in Gqeberha in Winter? (1 x 2) (2)

[Concept]

Increased pollution reacts with water to form more acidic rain ✓✓

10.2.2. Pollution domes are a major concern for this area. Draw a simple diagram showing the formation of a pollution dome. (2 x 1) (2)



10.2.3. Suggest ONE way that the people of Gqeberha can combat the effects of pollution domes. (1 x 2) (2)

Greening, fines, laws, education etc. ✓✓

[12]

QUESTION 11: GIS

Refer to the orthophoto and answer the questions that follow.

11.1. Is the orthophoto an example of raster or vector data? (1 x 1) (1)

Raster ✓

11.2. What is the feature found between points 6 and 7? (1 x 1) (1)

Runway ✓

11.3. Buffering was clearly used before building this feature. Provide evidence from the map to justify this statement. (1 x 1) (1)

Lack of vegetation/settlement near here ✓

No mountains or other obstructions around ✓

Very flat area identified ✓

11.4. Why would we need to use buffering before building the feature identified in QUESTION 11.2? List TWO reasons. (2 x 1) (2)

[Accept reasonable answers]

Lands needs to be flat ✓✓

Noise pollution is bad for animals and humans ✓✓

A large area will be affected and so we need to identify endangered species etc. ✓✓

Water levels will increase maintenance costs ✓✓

Tourist usage and viability needs to be considered ✓✓

11.5. Wind patterns are important for the safe functioning of the feature identified in QUESTION 11.2. Would GIS have been used to study the wind patterns before construction? (1 x 1) (1)

Yes ✓

11.6. How would the GIS wind data have been used in this construction? (1 x 2) (2)

Aircraft should not take off against prevailing winds ✓✓

Aircraft can use the winds to increase speed and aid with take-off ✓✓

Need to understand the swirls when landing ✓✓

[8]

TOTAL SECTION C: [30]

GRAND TOTAL: 150 Marks